Supporting Information

Effects of Initial Crystal Structure of Fe_2O_3 and Mn Promoter on Effective Active Phase for Syngas to Light Olefins

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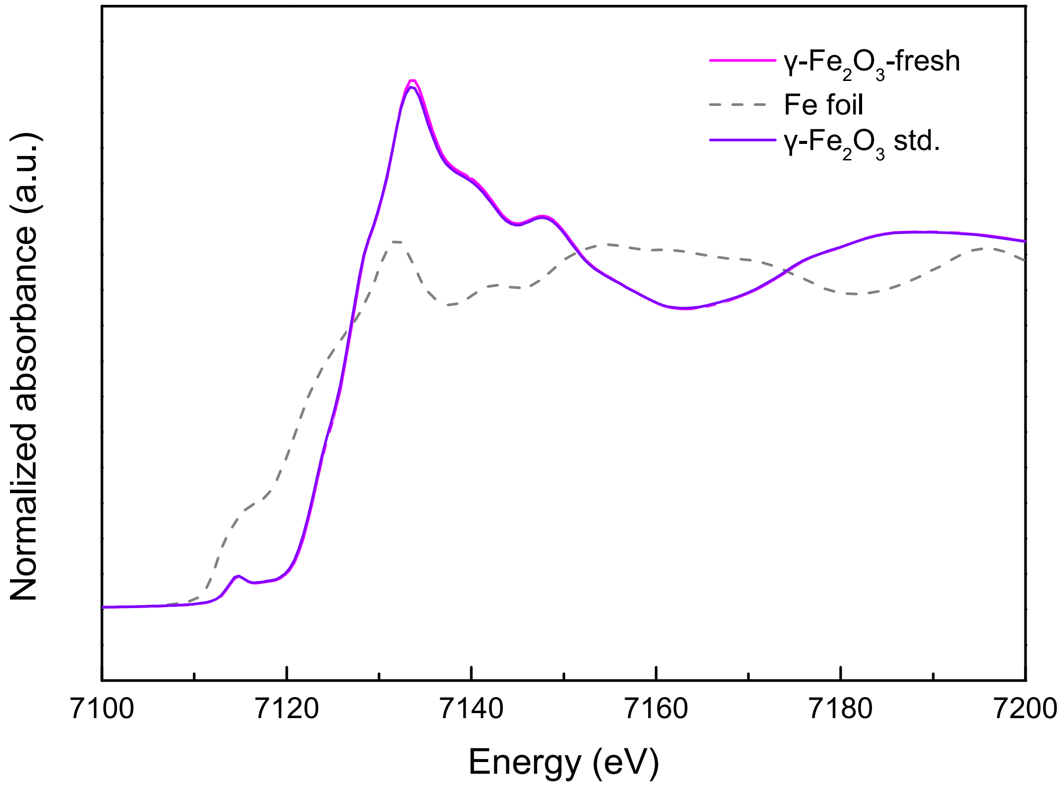

Figure S1. Fe K-edge XANES of fresh γ-Fe_2O_3 catalyst and Fe standards.

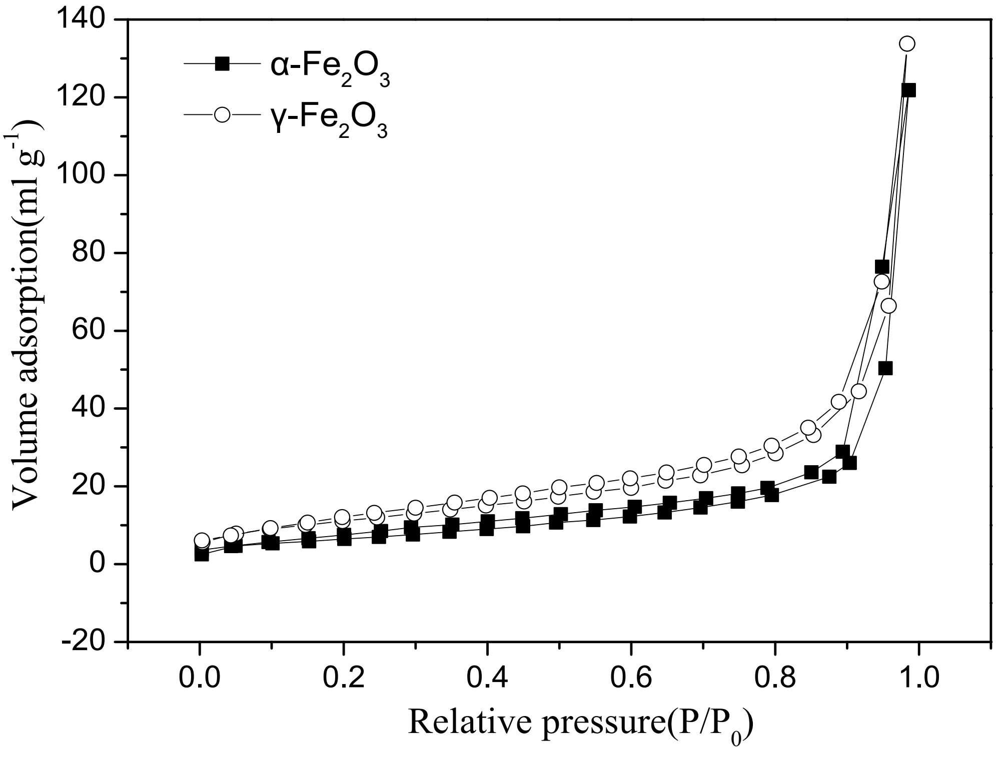

Figure S2. N_2 adsorption-desorption isotherms of α-Fe_2O_3 and γ-Fe_2O_3 nanorods as prepared.

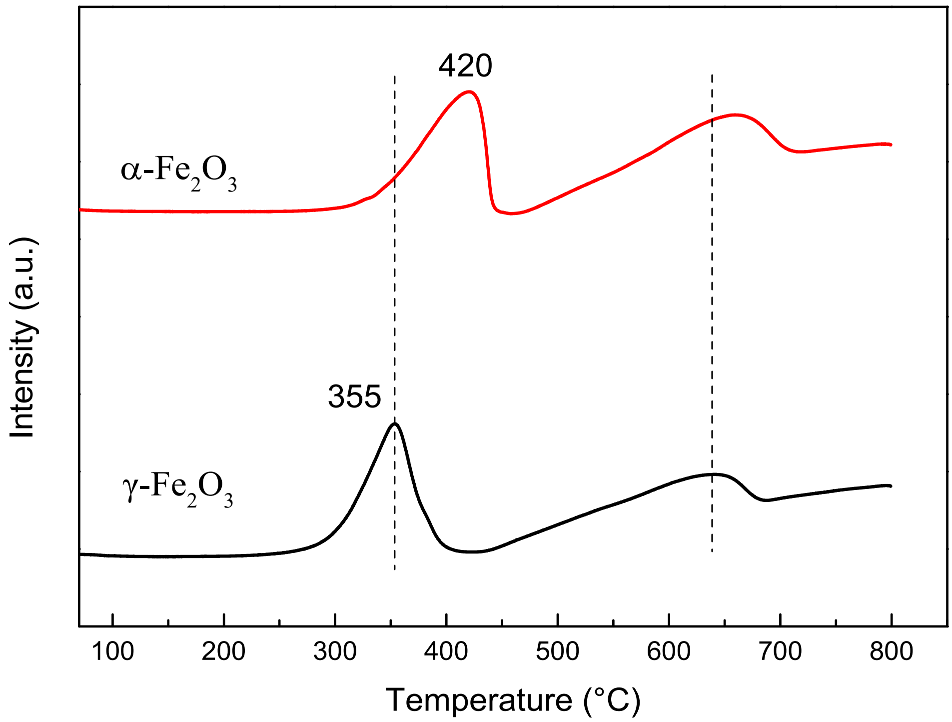

Figure S3. H_2-TPR profiles of α-Fe_2O_3 and γ-Fe_2O_3 nanorods catalysts.

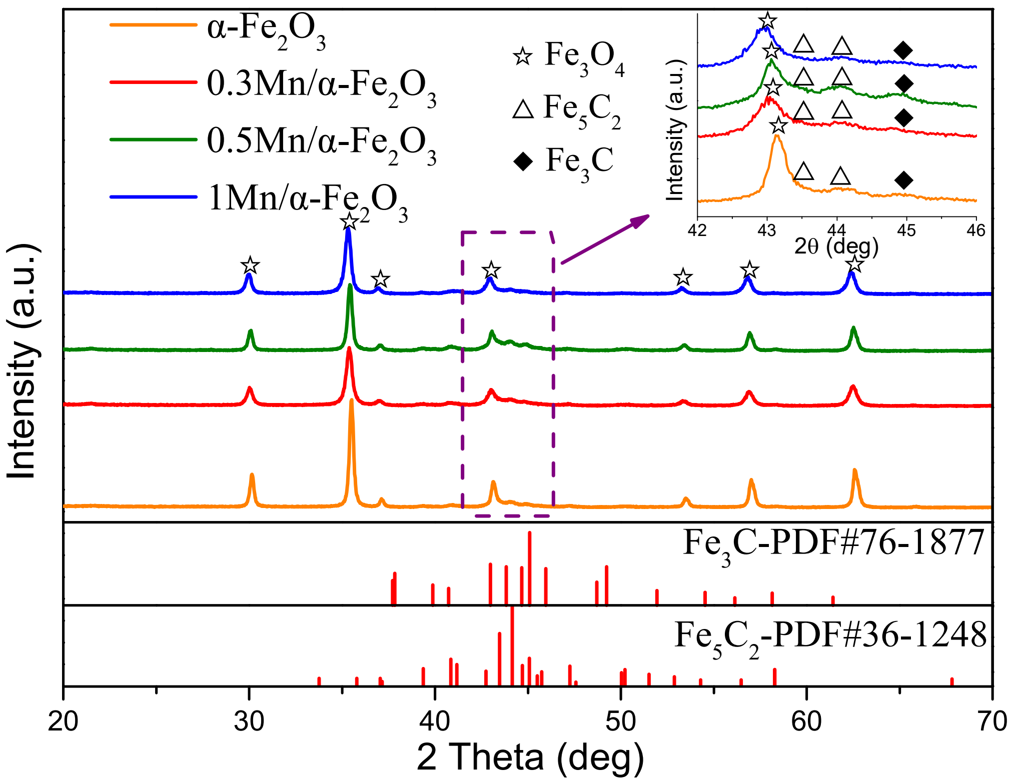

Figure S4. XRD patterns for the α-Fe_2O_3 nanorods and Mn promoted samples after carburization.

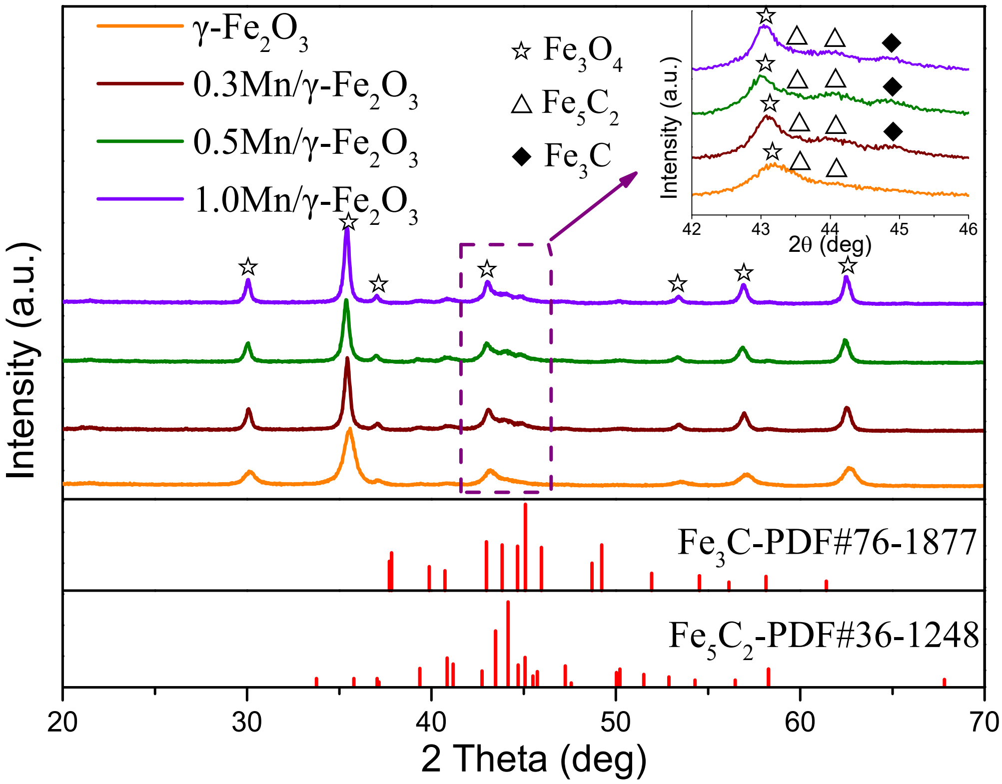

Figure S5. XRD patterns for the γ-Fe_2O_3 nanorods and Mn promoted samples after carburization.

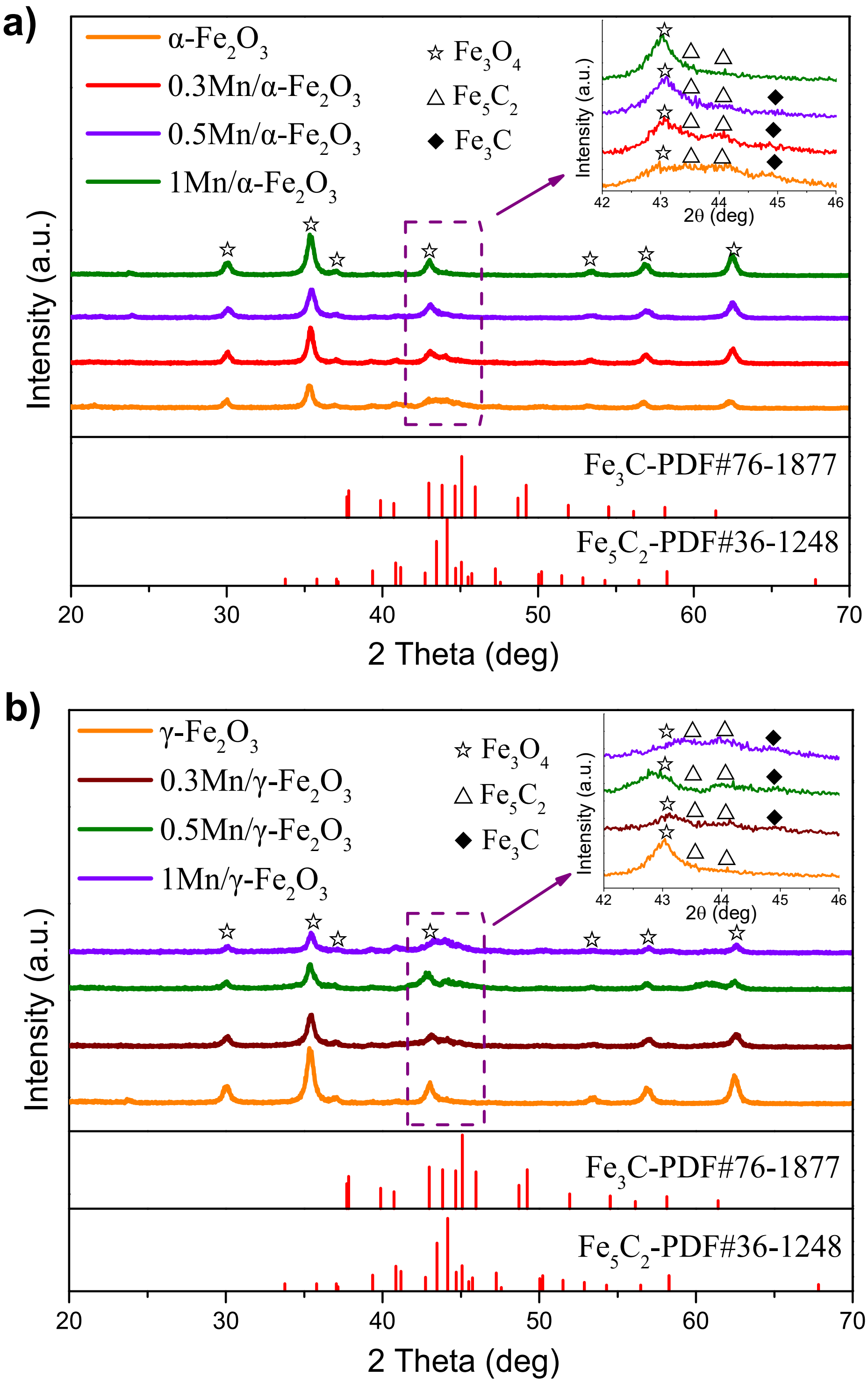

Figure S6. a) XRD patterns of α-Fe_2O_3 nanorods and Mn promoted samples after 20 h reaction. b) XRD patterns of γ-Fe_2O_3 nanorods and Mn promoted samples after 20 h reaction.

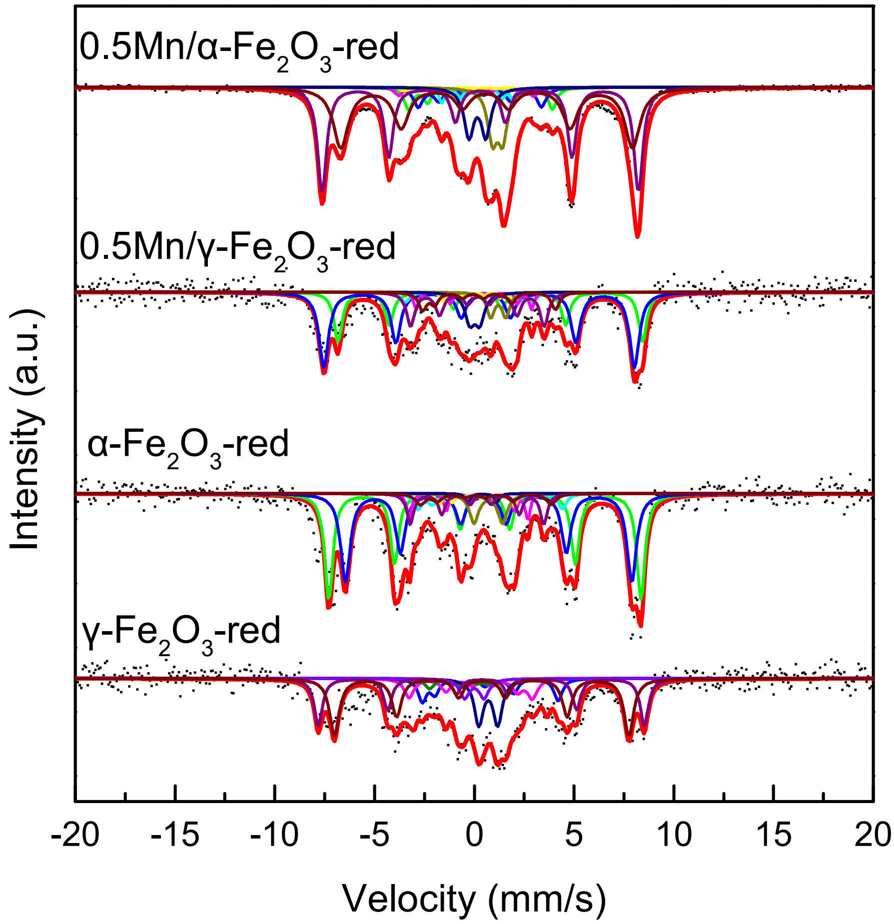

Figure S7. Mössbauer spectra of α-Fe_2O_3 and γ-Fe_2O_3 nanorods catalysts after carburization. Carburization conditions: 340 ºC, 0.1MPa, H_2/CO=1, W/F(CO+H_2+Ar) =2.5 g_{cat}·h·mol^{-1}.

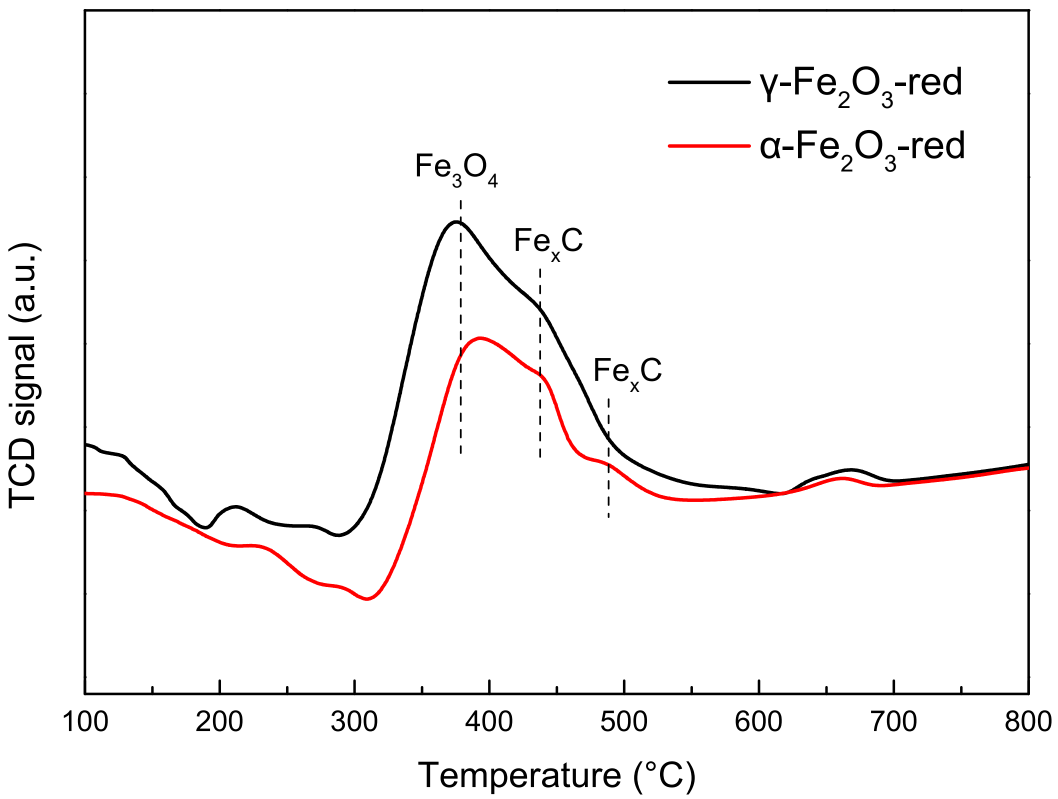

Figure S8. CO-TPD profiles of α-Fe_2O_3 and γ-Fe_2O_3 nanorods catalysts after carburization.

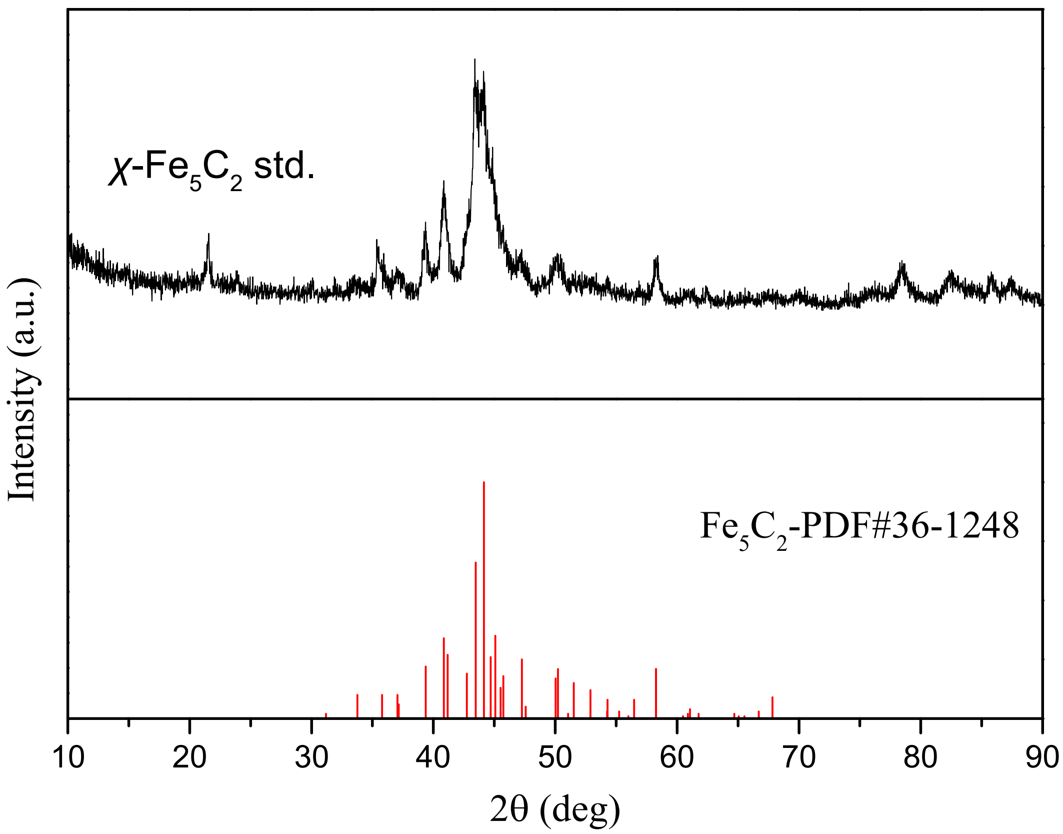

Figure S9. XRD spectra of χ-Fe_5C_2 reference sample as prepared by in situ carbonization of hematite.

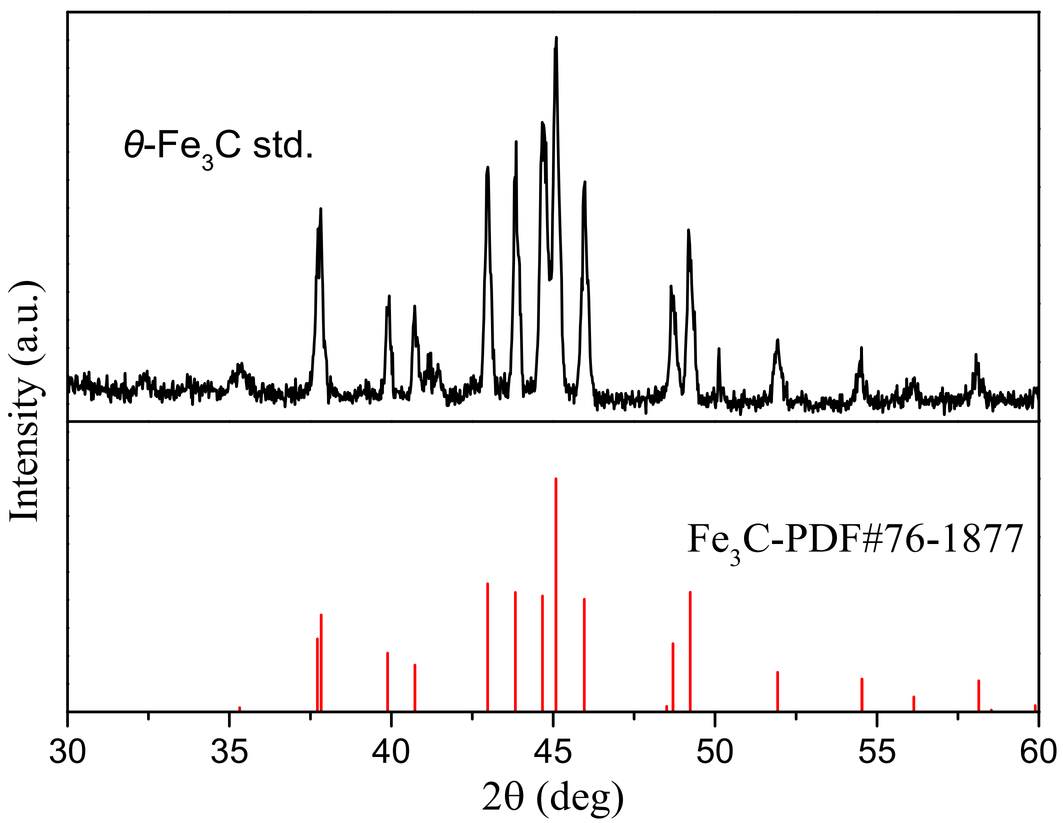

Figure S10. XRD spectra of θ-Fe_3C reference sample as prepared by wet-chemical route.

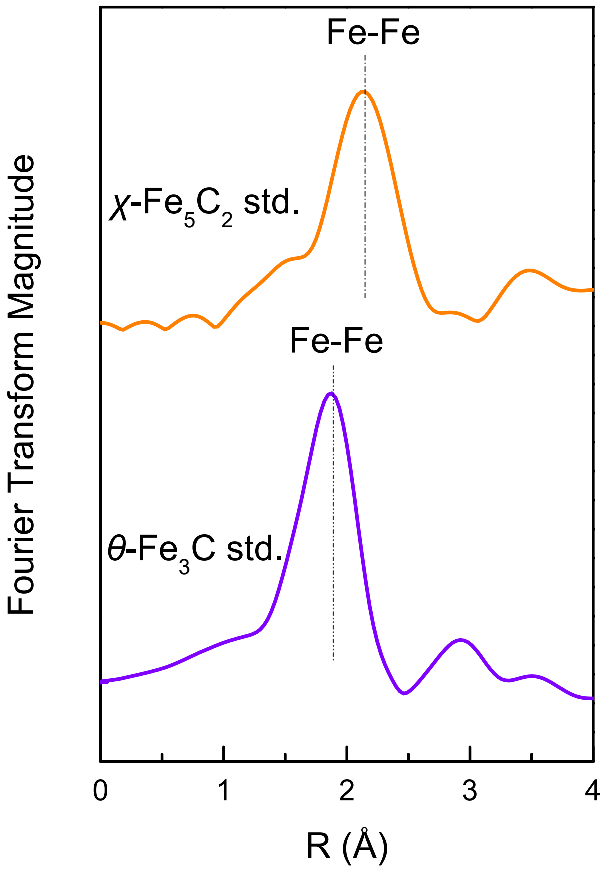

Figure S11. The nonphase corrected, Fourier transformed (FT) Fe K-edge EXAFS spectra of χ-Fe_5C_2 and θ-Fe_3C reference samples.

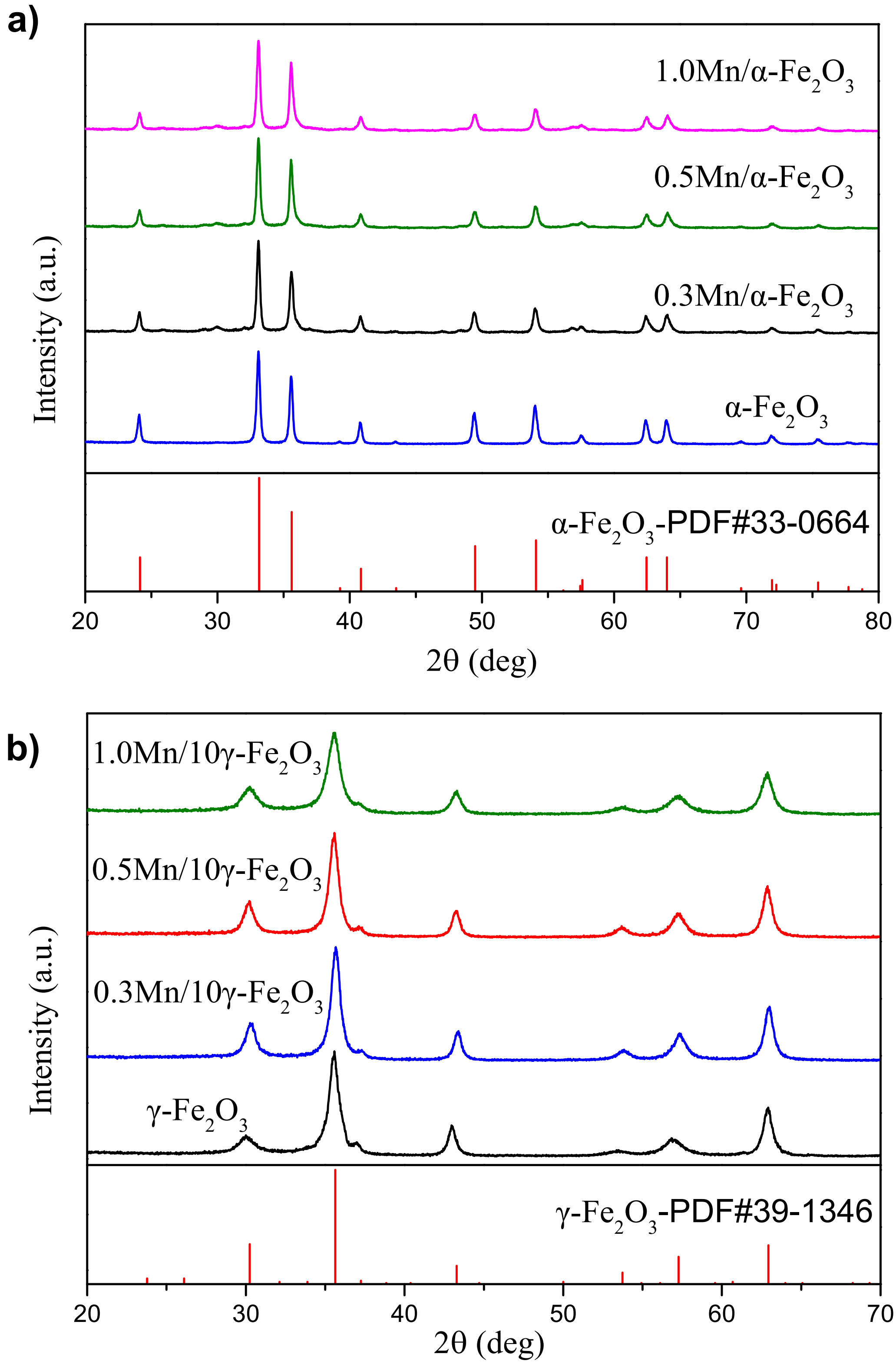

Figure S12. X-ray diffraction of α-Fe_2O_3 (a) and γ-Fe_2O_3 (b) nanorods catalysts with different manganese loading as prepared.

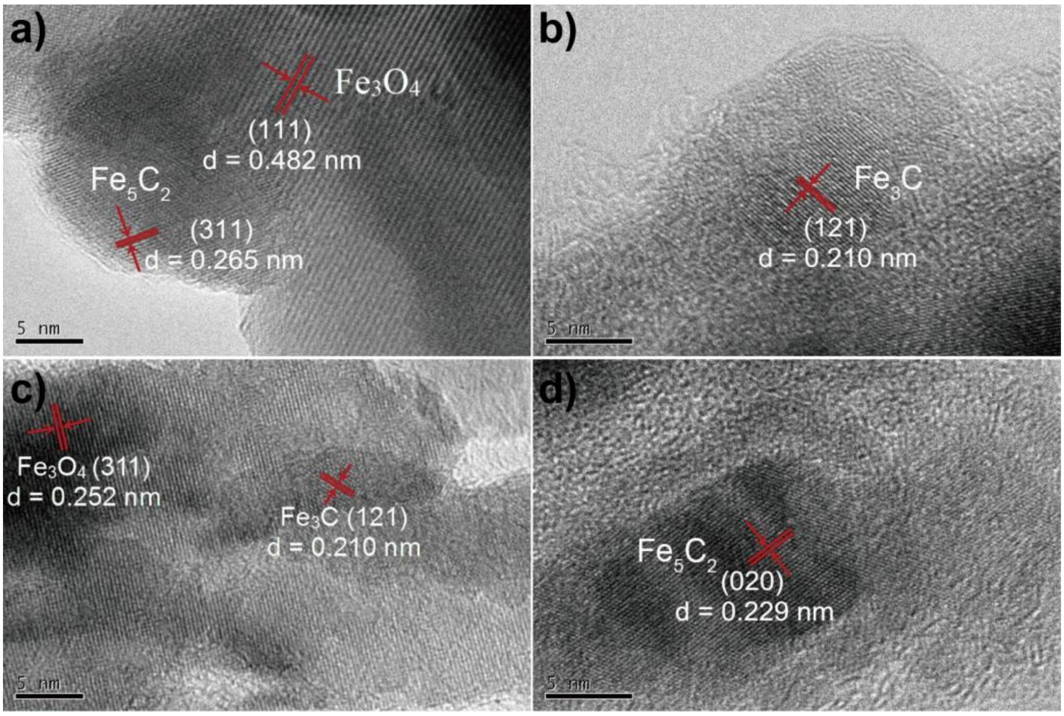

Figure S13. HRTEM of 0.5Mn/α-Fe_2O_3 (a, b) and 0.5Mn/γ-Fe_2O_3 (c, d) nanorods catalysts after carburization. Carburization conditions: 340 ºC, 0.1 MPa, H_2/CO=1, W/F(CO+H_2+Ar) =2.5 g_{cat}·h·mol^{-1}.

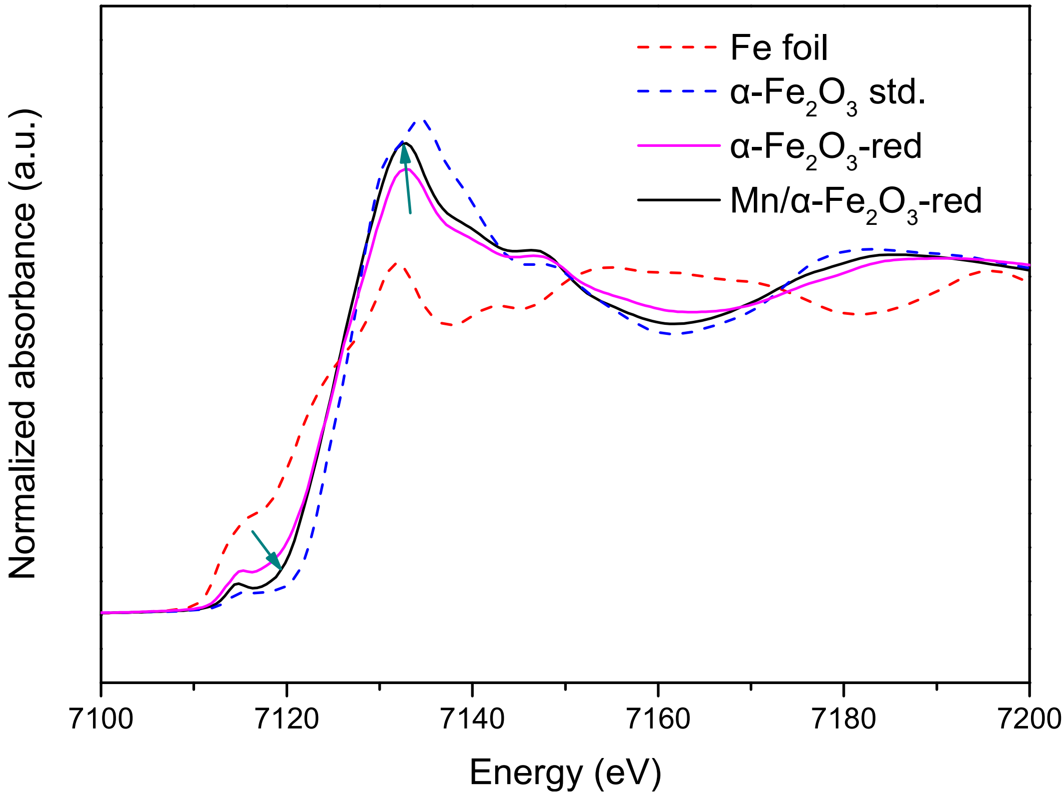

Figure S14. *In situ* Fe K-edge XANES of 0.5Mn/α-Fe_2O_3 catalyst and Fe standards. The pre-carburized catalysts were carburized again in-situ in syngas (CO/H_2=1) at 340 ºC for 1 h.

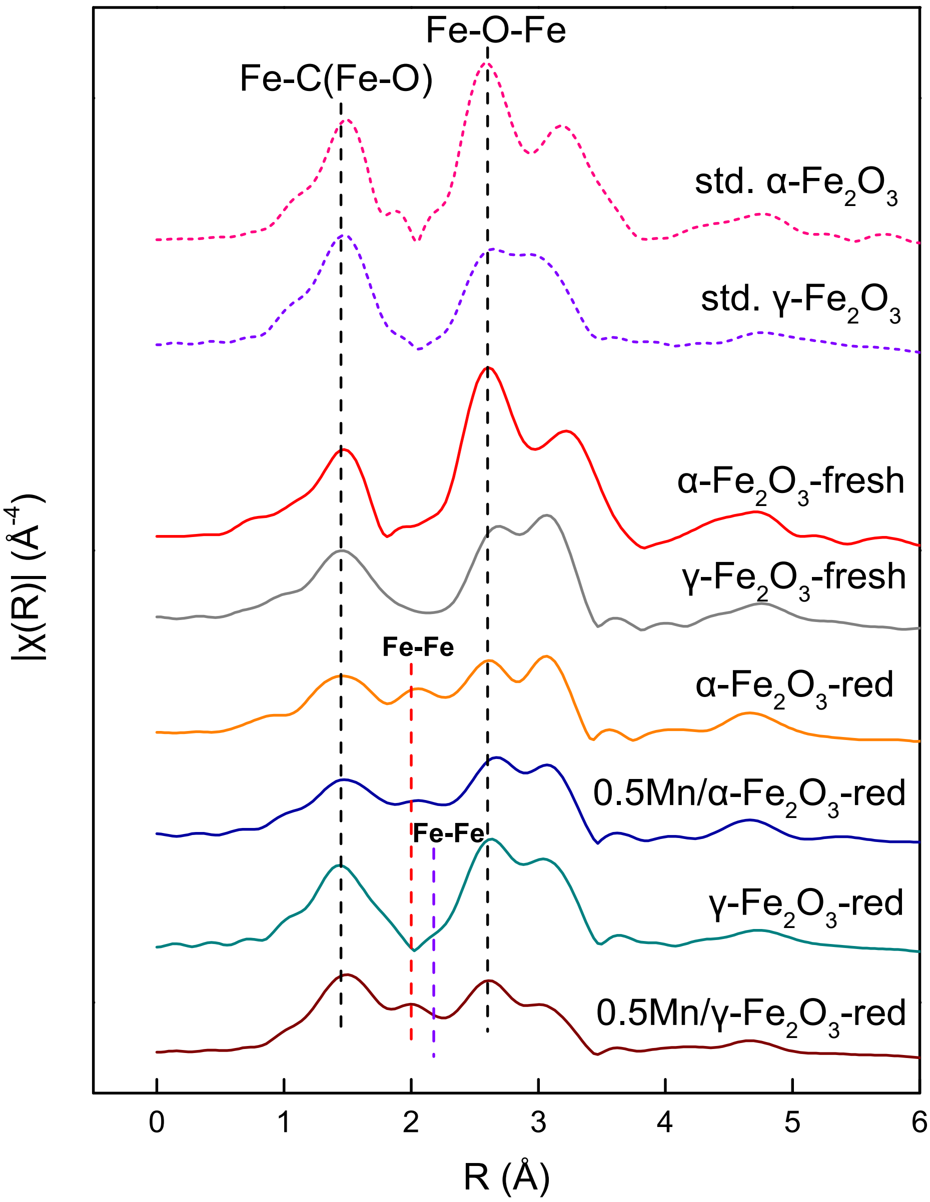

Figure S15. The nonphase corrected, k^3 weighted Fourier-transformed EXAFS spectra at the Fe K-edge of the α-Fe_2O_3 and γ-Fe_2O_3 based catalysts as-prepared and after carburized. Dash lines denote reference samples of α-Fe_2O_3 and γ-Fe_2O_3, and solid lines denote experiment spectra of iron based catalysts. The carburized samples were in situ treated in syngas (H_2/CO=1, 15 ml/min) at 340 ºC for 2 h at 1 bar.

Table S1 Least squares fitting of XANES data for various spent iron-based catalysts.

Catalysts	**Fe_2O_3**	**Fe_3O_4**	**Fe_xC[a]**
γ-Fe_2O_3	-	63.3 %	36.7 %
0.5Mn/γ-Fe_2O_3	-	56.1 %	43.9 %
α-Fe_2O_3	2.8 %	75.8 %	31.4 %
0.5Mn/α-Fe_2O_3	7.9 %	70.0 %	22.1 %

a Total content of Fe_5C_2+Fe_3C